卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-ENG-ECI-DD-GL-0102 30-B003701S-Z0102

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图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

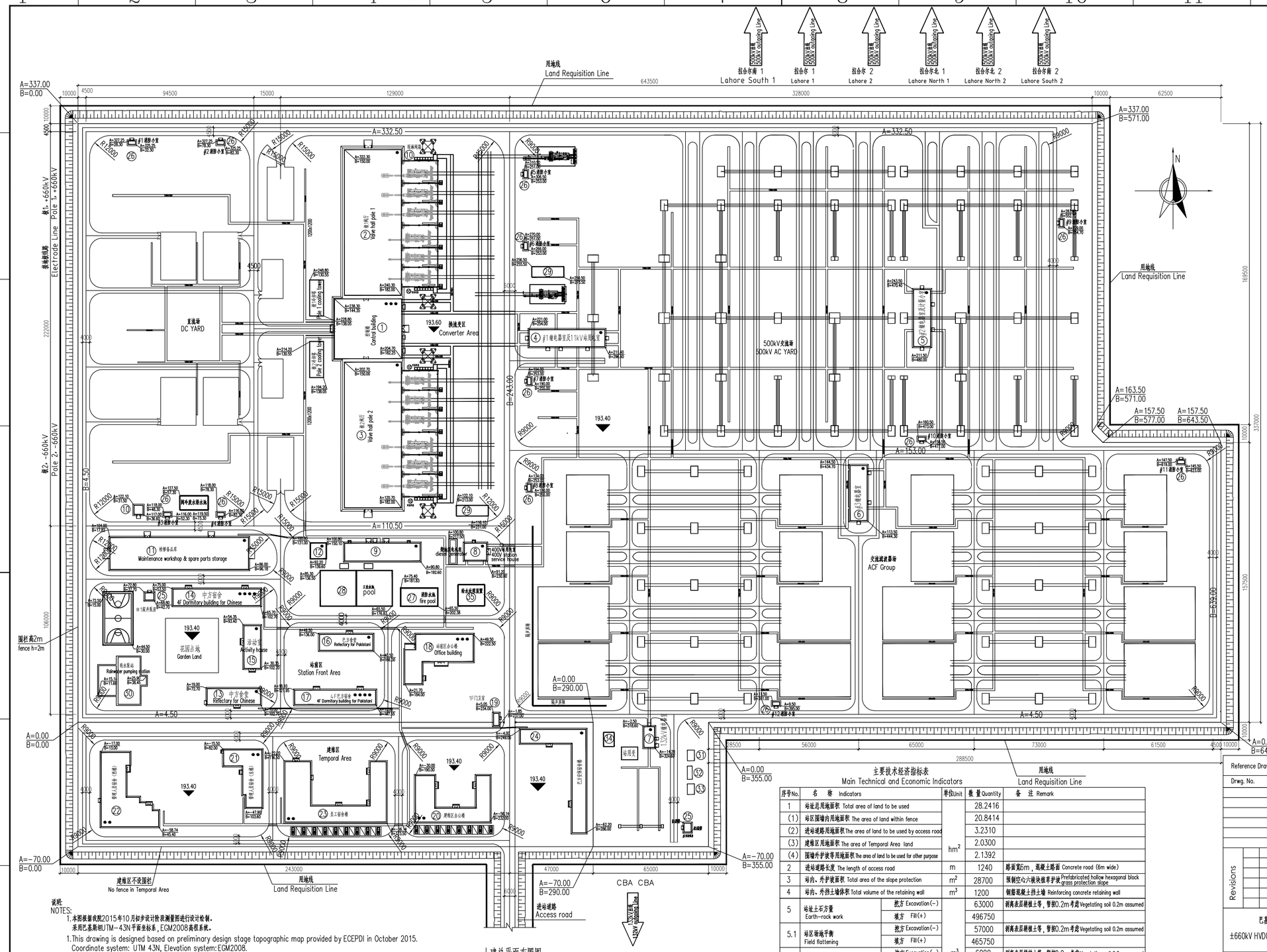
总图 专业 第 01 卷 第 02 册
General Layout SPECIALITY VOLUME BOOK

卷册名称 总平面布置及结构设计说明
Volume name General Layout and Specification of Structural Design

批 准 APPROVED BY	
审 核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	
图纸 2 张 DRAWING NOS.	打印件 0 本 PRINTED COPIES NOS.

序号 No.	图 号 DRAWING NO.	文件名称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
1	30-B003701S-Z0102-001	土建总平面布置图 General Layout and Specification of Structural Design	A	1	0	
2	30-B003701S-Z0102-002	土建结构设计总说明 General Description of Civil Structure Design	A	0	1	

注:



说明:
NOTES:

1.本图根据设院2015年10月初步设计阶段测量图进行设计绘制。
采用巴基斯坦UTM-43N平面坐标系，EGM2008高程系统。
1.This drawing is designed based on preliminary design stage topographic map provided by ECEPDI in October 2015.
Coordinate system: UTM 43N, Elevation system:EGM2008.

2.平面坐标以米为单位，尺寸以毫米为单位。
2.The plane coordinates are in meters and the dimensions are in millimeters.

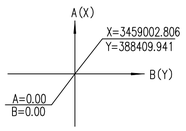
3.换流站场地平均设计标高为193.40m，换流变广场平均设计标高为193.60m。
3.The average design site level is equal to 193.40m.

4.站内道路为郊区型道路，道路中心标高193.60m。
4.Interior Road is suburban type, the center level of interior road is 193.60m.

5.未注明的道路宽度均为3.0米，道路的转弯半径一般均为7.0米。
5.The width of interior roads which are not indicated is 3.0m.
The average turning radius of interior roads is 7.0m.

6.本图建筑(A、B)坐标与UTM-43N平面坐标系关系如下所示:
6.The relationship between relative coordinate system and UTM-43N system is as below:

7.进站道路路面宽6.0m，自站区南侧接入Bhai Pheru-Mor Khunda Road公路，长度约1240m。
7.The width of access road is 6.0m. The access road connects to the Bhai Pheru-Mor Khunda Road highway. And the length of it is about 1240m.



主要技术经济指标表
Main Technical and Economic Indicators

序号No.	名 称 Indicators		单位Unit	数 量Quantity	备 注 Remark
1	站址总用地面积 Total area of land to be used		hm^2	28.2416	
(1)	站区围墙内用地面积 The area of land within fence			20.8414	
(2)	进站道路用地面积 The area of land to be used by access road			3.2310	
(3)	建设区用地面积 The area of Temporal Area land			2.0300	
(4)	围墙外护坡等用地面积 The area of land to be used for other purpose			1.2392	
2	进站道路长度 The length of access road		m	1240	路面宽6m，混凝土路面 Concrete road (6m wide)
3	站内、外护坡面积 Total area of the slope protection		m^2	28700	预制空心六棱块植草护坡 Prefabricated hollow hexagonal block grass protection slope
4	站内、外挡土墙体积 Total volume of the retaining wall		m^3	1200	钢筋混凝土挡土墙 Reinforcing concrete retaining wall
5	站址土石方量 Earth-rock work	挖方 Excavation(-)	m^3	63000	剥离表层耕植土等，暂按0.2m考虑Vegetating soil 0.2m assumed
		填方 Fill(+)		496750	
5.1	站区场地平衡 Field flattening	挖方 Excavation(-)		57000	剥离表层耕植土等，暂按0.2m考虑Vegetating soil 0.2m assumed
		填方 Fill(+)		465750	
5.2	进站道路 Access road	挖方 Excavation(-)		6000	剥离表层耕植土等，暂按0.2m考虑Vegetating soil 0.2m assumed
		填方 Fill(+)		31000	
5.3	站址土方综合平衡后需 After earth-rock volume is balanced	弃土 Spoil		42000	考虑耕植土2.1万方还还利用 Consider returning 21,000 soil for use
		取土 Borrow		546800	其中5.7万方于处平 50,000 of them are used for final level
6	站内道路面积 The area of internal roads		m^2	29860	混凝土路面 Concrete road
7	主电缆沟长度 The length of main cable trench in substation		m	6700	0.6mX0.6m及以上 0.6mX0.6m and above
8	总建筑面积 Total building area		m^2	26493.28	
9	站区围栏长度 The length of fence		m	2100	高度2.0m，钢丝网围栏 Wire fence, h=2m
10	屋外配电装置场地处理面积 The area of paved floors in outdoor switchgear yard		m^2	105000	堆卷走寿石碴3.7灰土150mm，再铺150mm碎石 Cover 3.7 gray soil 150mm and then cover 150mm gravel
11	换流变搬运轨道长度 The length of rail track		m	2300	

建(构)筑物一览表
LIST OF BUILDINGS AND STRUCTURES IN STATION

编号 No.	名 称 Name	建筑面积(m^2) Buildings area	占地面积(m^2) Land area	备 注 Remark
①	控制楼 Control building	3243	1309	三层框架结构 Reinforcing frame structure (3F)
②	极1阀厅 Pole 1 Valve Hall	2795	2795	单层钢结构 Steel structure (1F)
③	极2阀厅 Pole 2 Valve Hall	2795	2795	单层钢结构 Steel structure (1F)
④	#1继电器室及1 1kV站用电室 500 kV No.1 Relay and 11kV Station Service House	421.02	421.02	单层框架结构 Reinforcing frame structure (1F)
⑤	#2继电器室及计量小室 500 kV No.2 Relay and Metering House	316.8	316.8	单层框架结构 Reinforcing frame structure (1F)
⑥	户内500kV继电器室 500 kV No.3 Relay-house	306.99	306.99	单层框架结构 Reinforcing frame structure (1F)
⑦	132kV继电器室及计量小室 132kV Relay House and Metering House	74.32	74.32	单层框架结构 Reinforcing frame structure (1F)
⑧	400V站用电室 400V Station Service House	125.14	125.14	单层框架结构 Reinforcing frame structure (1F)
⑨	综合泵房及消防泵房 Pump House	614.61	614.61	单层框架结构 Reinforcing frame structure (1F)
⑩	专用仓库 Special Materials Storage	28.41	28.41	单层框架结构 Reinforcing frame structure (1F)
⑪	维修备品库 Maintenance Workshop & Spare Parts Storage	1432.7	1432.7	单层砖结构 Rock structure (1F)
⑫	备用平波室 Standby Smoothing Reactor House	80.1	80.1	单层钢结构 Steel structure (1F)
⑬	中方食堂 Refectory for Chinese	279.02	279.02	单层框架结构 Reinforcing frame structure (1F)
⑭	中方宿舍 Dormitory Building for Chinese	1963.05	471.07	四层框架结构 Reinforcing frame structure (4F)
⑮	活动室 Activity House	230.91	230.91	单层框架结构 Reinforcing frame structure (1F)
⑯	巴方食堂 Refectory for Pakistani	279.02	279.02	单层框架结构 Reinforcing frame structure (1F)
⑰	巴方宿舍 Dormitory Building for Pakistani	1585.46	381.76	四层框架结构 Reinforcing frame structure (4F)
⑱	站前区办公楼 Office Building	1972.23	672.53	三层框架结构 Reinforcing frame structure (3F)
⑲	门卫室 Guard Room	23.03	23.03	单层框架结构 Reinforcing frame structure (1F)
⑳	建设区办公楼 Office Building	1395	697.41	二层砖砌结构 Masonry structure (2F)
㉑	管理人员宿舍(东楼) Dormitory Building for Managers(E)	947	447.18	二层砖砌结构 Masonry structure (2F)
㉒	管理人员宿舍(西楼) Dormitory building for Managers(W)	1607	869.55	二层砖砌结构 Masonry structure (2F)
㉓	员工宿舍楼 Dormitory Building for Staff	3118	1124.46	二层砖砌结构 Masonry structure (2F)
㉔	巴方安保宿舍楼 Dormitory Building for Security Personnel	703.19	703.19	单层砖砌结构 Masonry structure (1F)
㉕	深井泵房(2个) Deep Well Room	22.12	22.12	单层框架结构 Reinforcing frame structure (1F)
㉖	消防小室(12个) Fire Equipment Storage Room	9.42	9.42	单层框架结构 Reinforcing frame structure (1F)
㉗	消防水池 Fire pool		128	地下钢筋混凝土结构 Underground reinforced concrete structure
㉘	工业水池 Industry pool		784	地下钢筋混凝土结构 Underground reinforced concrete structure
㉙	换流变事故油池(2个) Accident oil pool		96	地下钢筋混凝土结构 Underground reinforced concrete structure
㉚	雨水泵站 Rainwater pumping station		600	地下钢筋混凝土结构 Underground reinforced concrete structure
㉛	污水调节池 Sewage control pool		24	地下钢筋混凝土结构 Underground reinforced concrete structure
㉜	污水处理装置 Sewage treatment plant		21	地下钢筋混凝土结构 Underground reinforced concrete structure
㉝	污水回用水池 Sewage back to pool		24	地下钢筋混凝土结构 Underground reinforced concrete structure
㉞	站用变事故油池 Accident oil pool		47	地下钢筋混凝土结构 Underground reinforced concrete structure
㉟	给水处理装置 Water treatment unit		50	
㊱	循环废水排水池 Vale-cooled wastewater pool		160	地下钢筋混凝土结构 Underground reinforced concrete structure
合计 Total		26493.28	18665.50	

图例:
Legend:

建筑物 Building
地下建筑 Building underground
架 构 Substaion frame
电缆沟 Cable trench
道 路 Road
道 坡 Slope
围 栏 Fence

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里-拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 SGCC 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - G L - 0 1 0 2 - 0 0 1 A

	姓名 Name	日期 Date
设计 Designed	郑理	20190515
校核 Checked	鲍林	20190515
审核 Reviewed	赵欣、杨柳青	20190515
批准 Approved	张安平	20190515
比例 Scale		

土建总平面布置图
General Layout and Specification of Structural Design

图号 DWG NO.: 30-B003701S-Z0102-001

版本 REV.: A

专业 Speciality	会签 Sign	日期 Date
变电一次	郑理	20190515

594X841

MLHVDC-LCS-ECI-DD-GL-0102-002A

30-B003701S-Z0102-002

巴基斯坦默蒂亚里-拉合尔±660kV 直流输电工程

拉合尔±660kV 换流站工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

±660kV Lahore Converter Station

施工图设计
Detail Design

总图部分

General Layout Part

土建结构设计总说明

General Description of Civil Structure Design

中国电力技术装备有限公司
China Electric Power Equipment and Technology Co., Ltd.
中国电力工程顾问集团华东电力设计院有限公司
East China Electric Power Design Institute of China Power Engineering Consulting Group

May, 2019

巴基斯坦默蒂亚里-拉合尔±660kV 直流输电工程

拉合尔±660kV 换流站工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

±660kV Lahore Converter Station

施工图设计
Detail Design

土建结构设计总说明
General Description of Civil Structure Design

批　准 Approved：

审　核 Reviewed：

校　核 Checked：

设　计 Designed：

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1 概 述

1.1 设计依据

(1) 初步设计文件
(2) 施工图阶段《岩土工程勘察报告》
(3) 施工图阶段各专业提资

1.2 场地概述

拉合尔换流站位于拉合尔市西南约 55km，站址的中心坐标约为：东经-73° 49'54"，北纬-31° 15'35"，位于 Bhai Pheru-Mor Khunda 公路西侧。站址地形平坦，自然地面标高约为 190.2-191.4m（大地水准面为 EGM2008，下同），场地大部分种植小麦，站址周边有多个敏感点：东侧距离一户农场主围院及其家禽养殖场约 170m，西侧距离 DHOPSARI VILLAGE 村庄最近处约 310m，距离坟场区约 80m；北侧距离 HASILMIR VILLAGE 村庄约 150m，南侧距离零散房屋约 450m。站址区域另有一条 500kV 线路需迁移。进站道路从 Bhar Pheru-Mor Khunda Rd 公路引接，进站道路长约 1.2km，该公路为沥青路面，路况良好，路面宽度约为 7m 左右，可以满足工程建设及大件运输要求。

1.3 工程地质和水文地质

1.3.1 工程地质

根据本次勘察成果，拟建站址场地位于河流冲积平原，在勘探深度范围内，整个站址区地层主要第四系全新统(Q_4^{al})冲积层组成，上部为粉质粘土，下部为粉细砂。

地基岩土层从上到下依次描述如下：

①层：粉质粘土，黄褐色、褐色，稍湿，以可塑状态为主，局部夹粉土，表层一般分布 30cm 厚耕植土。该层层位相对稳定，在整个场区广泛分布，厚度 2.0~7.0m，标贯击数 5.5 击，压缩模量 Es 值为 6.5MPa，地基承载力为 70kPa；

1 GENERAL

1.1 Design Basis

(1) Preliminary design document
(2) Geotechnical Engineering Survey Report
(3) Various professional materials

1.2 Overview of Station Sites

The site (Center Coordinate:N 31°15'35",E 73°49'54") is located at about 55km southwest of Lahore city and on the west side of Bhai Pheru-Mor Khunda highway. The site with flat terrain has a natural ground level of about 190 m to 191 m (geoidal surface is EGM2008, the same below). Wheat is planted on most of the site. The access road is connected from the Bhai Pheru-Mor Khunda highway, which is paved with bituminous surface, has a good condition and a width of about 7 m, and can meet the requirements for project construction and transportation of large equipment.

1.3 Engineering Geology and Hydrogeology Conditions

1.3.1 Engineering Geology Conditions

According to the survey results, the site is located in the river alluvial plain and within the scope of exploration depth. The stratum of the entire site mainly consists of Quaternary Holocene (Q_4^{al}) alluvial layers, the upper part is silty clay and the lower part is silty sand.

The rock and soil layers of foundation are described as follows from top to bottom:

①Layer: Silty clay, tawny, brown, slightly wet, mainly in a plastic state, locally interbedded with silt. 30cm thick cultivated soil is generally distributed on the surface layer. The layer has a relatively stable position and is widely distributed in the entire site, with the thickness of 2.0~7.0m, standard penetration test(SPT) blow count of 5.5, the compression modulus Es of 6.5MPa and the foundation bearing capacity of 70kPa;

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②层：粉细砂，灰色、青灰色，饱和，含有少量云母片，局部见砾石，夹少量粉土和中砂，分为②$_1$、②$_2$、②$_3$、②$_4$和②$_5$五个亚层；

②$_1$ 粉细砂：松散，层厚 2.0～7.0m，标贯击数 7.8 击，压缩模量 Es 值为 5.0MPa，地基承载力为 80kPa，该层在拟建场地内广泛分布；

②$_2$ 粉细砂：稍密，层厚 3.0～7.0m，标贯击数 12.6 击，压缩模量 Es 值为 7.0MPa，地基承载力为 100kPa，该层在拟建场地内广泛分布；

②$_3$ 粉细砂：中密，层厚 6.5~16.0m，标贯击数 22.2 击，压缩模量 Es 值为 15.0MPa，该层在拟建场地内广泛分布；

②$_4$ 粉细砂：密实，层厚 10.0~18.5m，标贯击数 35.2 击，压缩模量 Es 值为 24.0MPa，该层在拟建场地内广泛分布；

②$_5$ 粉细砂：极密，标贯击数大于 50 击，压缩模量 Es 值为 30.0MPa，在站址区均有分布，本次勘测未揭穿该层。

1.3.2 地震

根据《巴基斯坦建筑规范》(2007)，换流站接地极属于 ZONE 2A 区，地震区影响系数 Z=0.15，场地土属于中软土，场地土类型为 SD 类，地震系数 Ca=0.22，Cv=0.32，根据规范计算得到特征周期 TS 等于 0.582s；根据巴基斯坦地面峰值加速度图，50 年超越概率为 10%的地面峰值加速度为 0.14g。

② Layer: Silty-fine sand, gray, green gray, saturated, containing a small amount of mica, locally with gravel, with a small amount of silt and medium sand. The layer is divided into five sublayers as ②$_1$,②$_2$,②$_3$,②$_4$ and ②$_5$;

②$_1$silty-fine sand is loose, with layer thickness of 2.0~7.0m, SPT blow count of 7.8, compression modulus Es of 5.0MPa, and foundation bearing capacity of 80kPa. The layer is widely distributed in the site;

②$_2$Silty-fine sand is slightly dense, with layer thickness of 3.0 ~ 7.0m, SPT blow count of 12.6, compression modulus Es of 7.0 MPa, and foundation bearing capacity of 100kPa. This layer is widely distributed in the proposed site;

②$_3$Silty-fine sand is medium dense, with layer thickness of 6.5 ~ 16.0m, SPT blow count of 22.2, and compression modulus Es of 15.0 MPa. This layer is widely distributed in the site;

②$_4$Silty-fine sand is dense, with layer thickness of 10.0 ~ 18.5m, SPT blow count of 35.2, and compression modulus Es of 24.0 MPa. This layer is widely distributed in the site;

②$_5$Silty-fine sand is very dense, with SPT blow count of larger than 50, and compression modulus Es of 30.0MPa. This layer is evenly distributed in the site and was not exposed in this survey.

1.3.2 Earthquakes

According to Building Code of Pakistan (2007), the Lahore converter station is mainly located in ZONE 2A. The Peak Ground Acceleration (g) with 10% Probability of Exceedance in 50 years is 0.14g.

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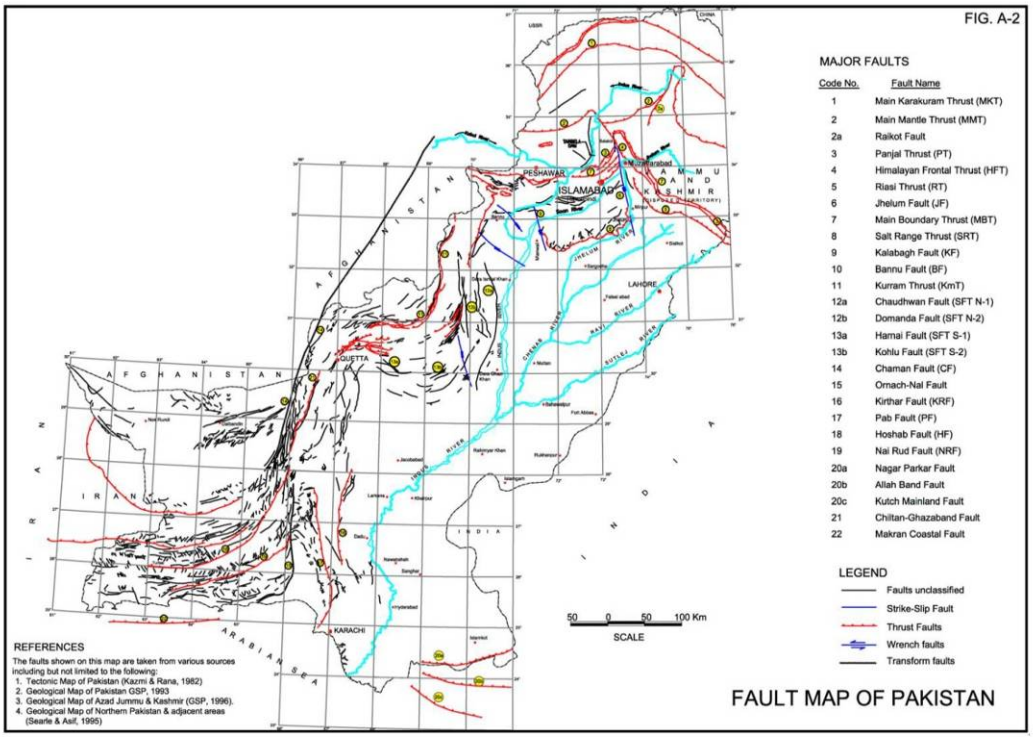

Fig. 1.3-1 Geologic Structure Map of Pakistan

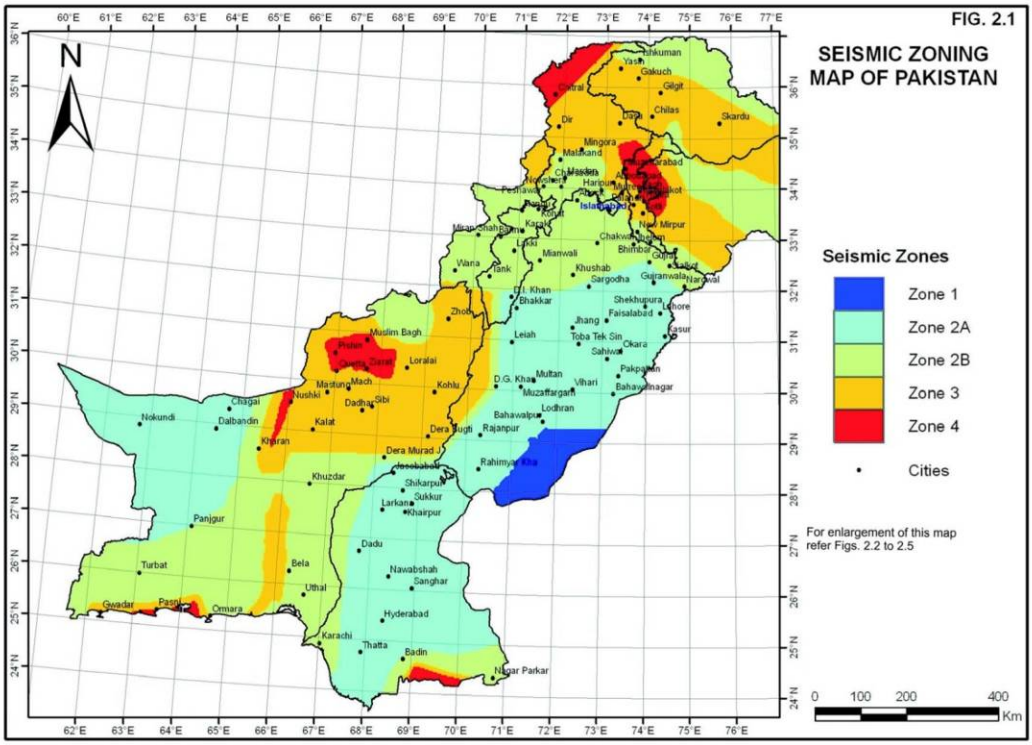

Fig. 1.3-2 Earthquake Partition Map of Pakistan

1.3.3 水文地质

根据场地地层分析，站址区地下水为潜水，潜水主要含水层为粉细，其赋水性较好，渗透性强，其补给来源主要为大气降雨、场区附近的沟渠，主要排泄渠道为大气蒸发和侧向补给给河

1.3.3 Hydrogeology

According to the analysis to formation of the site, the underground water in the site region is dive, and the basal water is fine-silty, with better water-bearing capacity, and strong penetration,

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流。本次勘测期间钻孔中的地下水稳定水位埋深为 3.32～4.87 米。根据抽水试验，地下水可满足换流站用水要求。

the supply source is mainly precipitation, channels near the site. In the period of exploration, the stable buried depth of underground water in the drilling hole is 3.32～4.87m. According to pumping test, the underground water can meet the water demands of the Lahore Converter station.

根据本次勘察水质及土的化学物质分析成果，依据《Building Code Requirements for Structural Concrete and Commentary（ACI 318-14）》判定：地下水对混凝土结构的暴露类别为 S1，场地土对混凝土结构的暴露类别为 S0，对钢筋混凝土结构中钢筋的暴露类别为 C2。

According to ACI 318-14 Building Code Requirements for Structural Concrete and Commentary, the exposure category of groundwater to concrete structure is S1, and the exposed type of soil to concrete is S0, to the steel bars in the reinforced concrete structure is C2.

1.3.4 地震液化

本工程场地的地震基本烈度为 7 度，地面下存在饱和砂土和饱和粉土时，应进行液化判定。按较高地下水位埋深 2.0m 进行考虑，根据美标 Seed 法（SPT）计算结果，本工程场地土在 7 度地震时，②$_1$、②$_2$ 粉细砂可发生液化，液化没有明显的区域性，一般 12.0m 深度内的粉细砂可能发生液化，液化等级中等。

1.3.4 Liquefaction

The converter station is located in the section of alluvial plain; there is saturated sand, with the risk of sandy soil liquefaction. The determination of this sandy soil liquefaction adopts American-standard of seed method (SPT). from the calculation results, there is sandy soil liquefaction in ②$_1$、②$_2$ soil layer of this site.

1.3.5 不良地质

根据液化判定结果，拟建场地内存在液化土层。除此之外，站址区及附近未发现滑坡、崩塌、岩溶、地面隆起沉陷、采空区、泥石流等不良地质作用。

1.3.5 Unfavorable Geologic Process

According to the liquefaction determination results, the site exists liquefiable soil layer. There isn’t other unfavorable geologic process in the Converter station region, such as landslide, collapse, mud-rock flow and ground settlement etc.

1.4 水文气象条件

拉合尔降雨较为充沛，大部分降雨集中在 7、8 月份。据现场调查，当发生夏季暴雨时，由于排水不畅可引起短时内涝，积水深度约 0.5m，内涝水位 191.8m，2~3 天即可排完。

1.4 Hydro-meteorological conditions

The rainfall in Lahore is rich and most concentrates in July and August. According to the site investigation, in case of rainstorm in summer, the poor discharge conditions may result in the short-period water logging with the water level of 191.8 m and the about 0.5m-deep ponding, which can be discharged in 2-3 days.

由历年最大洪水资料，进行 PIII 型频率计算后，推算 100 年一遇洪水位为 193.23m。

Based on the largest flood data in history, the P-III frequency is calculated, and the 100-year flood level is calculated to be 193.23 m.

拉合尔气象站常规气象资料如下（除个别值外，资料统计年限为 1980-2010）：

The general meteorological data of the Lahore Weather Station are listed below (the data was

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年平均气温 23.9℃
极端最高气温 48.0℃ (June 10, 2007)
极端最低气温 -2.2℃ (January 22, 2008)
年平均气压 983.1 hPa
平均降水量 679.7 mm
24 小时最大降雨量 274 mm (September 6, 2014)
年平均蒸发量 1,623.6 mm
年平均相对湿度 57%
年平均风速 1.6 m/s
实测最大风速 40.7 m/s (3s 瞬时)
主导风向 NW

结合本工程前期会议纪要，本工程设计风速采用 160km/h（对应风速约为 31.1m/s），设计覆冰厚度为 0mm。

1.5 主要建筑材料

（1）现浇钢筋混凝土结构及基础
混凝土： fc'=21MPa、fc'=28MPa
钢 筋： ASTM A615：Grade 60、Grade 40
（2）砌体结构
砌块： f'm=15Mpa（2200psi）
商品砂浆： f'g=17.2Mpa（2500psi）
（3）钢结构
钢材： Q235B、Q345B
螺栓： 6. 8 级、8. 8 级

2 站区总布置与交通运输

2.1 站区总体规划

拉合尔换流站位于拉合尔市西南约 55km，Bhai Pheru-Mor Khunda 公路西侧。站址地势平坦，自然地面标高约为 190.2～191.4m（大地水准面为 EGM2008，下同），场地大部分种植小麦。

collected from 1980 to 2010 except for partial data):
Annual mean air temperature 23.9℃
Extreme maximum air temperature 48.0℃ (June 10, 2007)
Extreme minimum air temperature -2.2℃ (January 22, 2008)
Annual mean atmospheric pressure 83.1 hPa
Mean rainfall 79.7 mm
Maximum rainfall in 24 hours 274 mm (September 6, 2014)
Annual mean evaporation capacity 1,623.6 mm
Annual mean relative humidity 57%
Annual mean wind speed 1.6 m/s
Measured maximum wind speed 40.7 m/s (3 s instant)
Prevailing wind direction NW

According to the minutes of the meeting held before, the design wind speed for this project is taken as 160 km/h (corresponding to about 31.1 m/s), and the design icing thickness is taken as 0 mm.

1.5 Materials

（1）RC structure and foundation
Concrete: fc'=21MPa、fc'=28Mpa
Steel bar: ASTM A615：Grade 60、Grade 40
（2）Masonry structure
Block: f'm=15Mpa（2200psi）
Mortar: f'g=17.2Mpa（2500psi）
（2）Steel structure
Steel: Q235B、Q345B
Bolt: Grade 6. 8、8. 8

2 GENERAL LAYOUT AND TRANSPORTATION

2.1 Overall Planning

The site (center coordinate: N-31°15'35", E73°49'54") is located at about 55km southwest of Lahore city and on the west side of Bhai Pheru-Mor Khunda highway. The site terrain is flat and its natural ground level is about 190 m to 191 m (geoidal surface is EGM2008, the same below).

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根据电气工艺规划，换流站 500kV 向北方向出线，直流与接地极向西方向出线，132kV 向南方向出线。

站址布置时，综合考虑电气总平面布置、线路进出线走向、站址自然条件、周围环境以及巴方提供的用地范围，确定站区采用正北方向布置。进站道路自站区南侧接入 Bhai Pheru-Mor Khunda Road 公路，长约 1240m。站址范围内有 500kV 线路需改线、周边有养鸡场需搬迁。

According to the electrical process plan, the transfer station 500 kV to the North, DC and ground pole to the West, 132 kV to the South.

When the site is arranged, it is determined that the station area is arranged in the North direction, taking into account the general layout of the electric plane, the direction of the line entry and exit, the natural conditions of the site, the surrounding environment, and the land area provided by the Pakistani side.The entrance road connects to the Bhai Pheru-Mor Khunda Road from the South Side of the station area and is about 1,240 m long.Within the site area, there are 500kV lines that need to be rerouted and chicken farms around them need to be relocated.

2.2 站区总平面布置

根据电气布置及工艺的要求，站区总平面布置将控制楼、阀厅和换流变区域布置在站区中央，500kV 交流滤波器场地布置在站区东南侧；直流场布置在站区西北侧，向西出线；500kV 交流配电装置场地布置在站区东北侧，向北出线。高、低端换流变东侧场地布置运输轨道，便于换流变的搬运和检修，充分利用换流变运输通道的场地，减少占地和基础工程量。

2.2 General Layout

According to the requirements of electrical layout and process, the general layout of the station area will be arranged in the center of the station area, the valve hall and the converter area, and the 500kV AC filter site will be arranged on the southeast side of the station area; The direct flow field is arranged on the Northwest side of the station area and goes West; The 500kV AC distribution device site is arranged on the northeast side of the station area and goes North. The transportation track is arranged on the East side of the high and low end converter, which facilitates the handling and overhaul of the converter, and makes full use of the venues of the converter channel to reduce the area and basic engineering work.

辅助生产区主要布置在站区西南侧，辅助建筑主要有办公楼、综合泵房及消防泵房、400V 站用电室、检修备品库、备用平抗室、工业水池、消防水池、中方食堂、中方宿舍、巴方食堂、巴方宿舍及活动室等。为节省电缆，方便运行，#1 继电器室及 11kV 站用电室、#2 继电器室及计量小室、#3 继电器室、132kV 继电器室及计量小室分别布置在交流配电装置场地内和交流滤波器场地内。

The auxiliary production area is mainly located on the Southwest side of the station area, The auxiliary buildings mainly include office buildings, comprehensive pumping stations and fire pumping rooms, 400V station electrical rooms, maintenance and spare goods depots, spare resistance rooms, industrial pools, fire pools, Chinese canteens, Chinese dormitories, Pakistani canteens, and Pakistani dormitories. Activity rooms and so on. To save cable and

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facilitate operation, # 1 relay room and 11kV station electrical room, # 2 relay room and metering room, # 3 relay room, 132kV relay room and metering room are arranged in the AC distribution device site and AC filter site respectively.

建维区布置在站区南侧，进站道路西侧，主要有办公楼、管理人员宿舍楼、员工宿舍楼、巴方安保宿舍楼。建维区四周不设置维护结构，与站区之间采用钢丝网围栏隔开；巴方安保宿舍楼位于站区入口、进站道路东侧。站址各功能区相互独立，又相互协调，便于生产运作，又有利于站内人员及换流站的安全。

Temporal Area is located on the South Side of the station area and on the West side of the entrance road. There are mainly office buildings, management dormitory buildings, staff dormitory buildings, and Pakistani security dormitory buildings. There is no maintenance structure around the built-up area, and it is separated from the station area by a steel mesh fence; The Pakistani security dormitory building is located at the entrance of the station area and on the East side of the entrance road. The functional areas of the site are independent of each other and coordinate with each other to facilitate production and operation, and it is also conducive to the safety of personnel and transfer stations in the station.

换流站东西方向围墙长 643.50m，南北方向围墙长 407m（含建维区），站址围墙内占地面积 20.8414hm^2。

The East-West wall of the station is 643.50 m long, and the north-south wall is 407m long(including Temporal Area). The site wall covers an area of 20.8414 hm^2.

2.3 竖向布置

根据测量报告，换流站站址地势自然标高在 190.2-191.4 左右，地势平坦，站址自然坡率 <5%，所以竖向布置采用平坡式布置。

2.3 Vertical Arrangement

According to the measurement report, the natural elevation of the transfer station site is about 190.2-191.4, the terrain is flat, and the natural slope rate of the station site is & lt; 5 %, so the vertical layout adopts a flat slope arrangement.

根据水文气象勘测报告，百年一遇设计洪水位为 193.23m，高于场地自然标高，因此需外购填土将换流站场地标高整体抬高至百年一遇设计洪水位以上。经初步分析，确定换流站场地设计标高取为 193.40m，综合考虑场地细平找坡后，确定各处竖向标高。

According to the hydrometeorological survey report, the design flood level in the first hundred years was 193.23 m, which was higher than the natural elevation of the site. Therefore, it is necessary to purchase the fill soil to raise the overall elevation of the exchange station site to above the design flood level in one hundred years. After preliminary analysis, it was determined that the height of the site design standard of the transfer station was 193.40 m.

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结合站址标高，站区初平土方总挖方量约 63000m^3，初平总填方量约 496750m^3。结合站内基础及沟道余土，并考虑耕植土 21000 m^3 返还利用，综合土方平衡后需弃淤泥及耕植土约 42000m^3，需外购土方约 546800m^3，其中 5 万 m^3 用于站址终平回填。

After considering the slope of the site, the vertical elevation of each place was determined. Combined with the elevation of the site, the total earth excavation volume in the station area is about 6300m^3, and the total fill volume is about 496750m^3. Considering the soil in the station foundation and ditch, after the comprehensive earthwork balance, it is necessary to abandon silt and cultivate about 42000m^3, and about 546800m^3 is borrowed to purchase earth.

站区内重要建筑物阀厅、控制楼室内地坪均高于换流变运输场坪 0.30m，换流变运输广场高于周边场地 0.20m。#1 继电器室及 11kV 站用电室、#2 继电器室及计量小室、#3 继电器室、132kV 继电器室及计量小室、办公楼、综合泵房及消防泵房等室内地坪高于周边室外地面 0.45m；道路中心标高高于周边场地标高 0.20m。

The main building valve hall and the control building floor in the station area are all higher than 0.30 m in the shunt transport yard, and the shunt transport Square is higher than 0.20 m in the surrounding venue. # 1 Relay Room and 11kV Station Room, # 2 Relay Room and Measurement Room, # 3 Relay Room, 132kV Relay Room and Measurement Room, Office Building, Integrated Pumping Room and Fire Pumping Room The floor is 0.45m above the surrounding outdoor ground; Road center elevation higher than surrounding site elevation 0.20m.

站区场地排水采用有组织排水，地面排水坡度 0.5%，坡向道路，道路两边每隔 20～30m 间距设置雨水井，各雨水井用管道相连通，最后接至雨水泵站，再通过雨水泵站集中排入至站外排水系统。

The station site drainage adopts organized drainage, the surface drainage slope is 0.5 %, and the slope is toward the road. The rain wells are set up every 20 to 30 meters on both sides of the road. The rain wells are connected by pipelines, and finally connected to the rain water pumping station, and then through the rain. Water pumping station is concentrated into the drainage system outside the station.

2.4 道路及管沟布置

站内道路根据环形畅通及换流变运输需要的原则，保证消防和有利生产的要求进行设置，其中换流变运输道路宽度 6.0m；平波电抗器运输道路宽度 4.5m，内转弯半径 15m；消防道路宽度 4.0m，转弯半径 9m；相间道路宽度 3.0m，转弯半径 7m。站区道路水泥混凝土路面，碎石基层路基。

2.4 Arrangement of Road and Underground Pipelines(Trenches)

Roads of cement concrete pavement will be used on the station site, and the roadbed will be made of macadam. Main roads (road for transport of converter transformers) on the site will be 6.0m wide. The roads for smooth reactors transportation will be 4.5m wide with a turning radius of 15m. The ring roads will be 4.0m wide with a turning radius of 9.0m. The

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phase-to-phase road will be 3.0m wide with a turning radius of 7.0m.

进站道路自站区南侧接入 Bhai Pheru-Mor Khunda Road 公路，长约 1240m；采用 6.0m 宽水泥混凝土路面，碎石基层，两侧各设置 0.5m 宽泥结碎石路肩。

The access road will be connected to the Bhai Pheru-Kot Radha Kishan Road in east. A 1240m long access road of 6.0m wide cement concrete pavement, gravel roadbed, and 0.5m wide hard shoulders on both sides will be built.

电缆沟均采用钢筋混凝土结构，预制混凝土盖板。

Cable trenches shall be of concrete structure.

Unpaved surface in the station shall be covered with gravel.

2.5 配电装置场地处理

根据电气要求，本工程配电装置场地采用碎石处理，喷药杀草后铺土工布，再上铺 200mm 碎石。站前区和建维办公区局部绿化。

2.5 Treatment of Outdoor Switchgear Yard

According to the electrical requirements, the outdoor switchgear yard is treated with gravel, after spraying the drug to kill the soil cloth, and then the upper 200 mm gravel. Local greening of the station front area and temporal area.

2.6 围栏及大门

结合业主要求，换流站区域采用 2m 高钢丝网围栏进行维护，建维区四周不设维护措施。进站大门采用钢丝网门；同时在进站大门入口处进行一定的艺术处理，使其与所前区建筑、绿化协调。

考虑站区安全需要及当地特殊情况，建议站区四周围墙（围栏）、岗楼等维护、安保设施具体布置要求需结合专业安保公司有关安保方案进行调整。

2.6 Fence and Gate

In accordance with the requirements of the owner, the transfer station area is maintained with a 2m high wire mesh fence, and no maintenance measures are provided around the construction area. The entrance gate adopts a steel wire mesh door; At the same time, a certain artistic treatment is carried out at the entrance of the entrance gate to make it coordinated with the construction and greening of the former area.

Taking into account the security needs of the station area and the special circumstances of the area, it is recommended that the maintenance of the perimeter walls(fences), guard buildings, etc., and the specific layout requirements of the security facilities of the station area be adjusted in conjunction with the relevant security programmes of the professional security companies..

3 结构设计说明

3.1 设计条件

设计风速：采用巴方推荐的 50 年一遇离地 10m 高瞬时(3s)最大风速分别为 160km/h，即 44.4m/s，换算成 10 分钟平均最大风速 31.3m/s。

3 DESCRIPTION OF STRUCTURE DESIGN

3.1 Design Conditions

Designed wind speed: the owner's recommended maximum instantaneous (3s) wind speed (at an altitude of 10m and frequency of once every 50

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根据地质调查，换流站属于 ZONE 2A 区，地震区影响系数 Z=0.15，场地土属于中软土，场地土类型为 SD 类，地震系数 Ca=0.22，Cv=0.32。根据规范计算得到特征周期 TS 等于 0.582s，根据巴基斯坦地面峰值加速度图，50 年超越概率为 10%的地面峰值加速度为 0.12~0.14g，相当于国内地震基本烈度为 7 度。

years) is 160km/h: 44.4m/s. It is converted into maximum average wind speed (10min) of 31.1 m/s.

Basic earthquake intensity: The converter station belongs to 2A area and the seismic zone factor Z=0.15. The soil profile type is SD, and the seismic coefficients Ca=0.22，Cv=0.32.The peak acceleration of ground motion is 0.14g under 10% frequency of once every 50 years.

3.2 设计标准

- ASCE 7-10 minimum design loads for buildings and other structures；
- ACI 318-14 Building Code Requirements for Structural Concrete and Commentary；
- AISC 341-05 seismic provision for structural steel buildings；
- ASCE 10-97 Design of Latticed Steel Transmission Structures；
- ASCE 113-2008 Substation Structure Design Guide；
- ACI 543R-12 Design, Manufacture, and Installation of Concrete Piles；
- AISC 360-10 Specification for Structural Steel Buildings；
- ACI 530-11 Building Code Requirements for Masonry Structures；
- Uniform Building Code（1997）。

3.2 Design Standards

- ASCE 7-10 minimum design loads for buildings and other structures.
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- AISC 360-10 Specification for Structural Steel Buildings.
- ACI 530-11 Building Code Requirements for Masonry Structures.
- Uniform Building Code (1997).

3.3 结构形式

3.3.1 建筑物

拉合尔站围墙内新建建筑物主要由三部分组成：

（1）生产区建筑：极 1 阀厅、极 2 阀厅、控制楼、#1 继电器室及 11kV 站用电室、#2 继电器室及计量小室、#3 继电器室、400V 站用电室、备用平波电抗器室、专用品库、综合泵房及消防泵房、检修备品库、132kV 继电器室等。

（2）站前区建筑：中方食堂、中方宿舍、

3.3 Structural Type

3.3.1 Buildings

±660kV Lahore converter station consists of the following three areas:

(1) Production Area: Valve hall pole1, Valve hall pole2, Control building, 500kV No.1 relay and 11kV station service power house, 500kV No.2 relay and metering house, 500kV No.3 relay house, 400V station service power house, Standby smoothing reactor house, Special materials storage, Pump house, Maintenance workshop & spare parts storage, 132kV relay house.

(2) Operation and Maintenance Area：Refectory

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活动室、巴方宿舍、巴方食堂、办公楼（站前区）、门卫室等。

for Chinese, Dormitory building for Chinese, Activity house, Refectory for Pakistani, Dormitory building for Pakistani, Office building (Operation and Maintenance Area), Guard house.

（3）建维区建筑：员工住宿、办公楼（建维区）、管理人员宿舍（东楼）、管理人员宿舍（西楼）、巴方安保宿舍等。

(3) Temporary building Area：Dormitory building for staff, Office building (Temporary building Area), Dormitory building for managers(east), Dormitory building for managers(west), Dormitory building for security personnel.

极 1 阀厅、极 2 阀厅主体结构采用单层钢结构，屋面采用梯形钢屋架形式；

Valve hall: PI valve hall and PII valve hall will be provided. The main structure is steel structure; the roof is in the form of trapezoid steel roof truss.

备用平抗室为单层门式刚架结构；检修备品库采用单层钢筋混凝土排架结构，屋架采用有檩屋盖结构体系；

Steel structure will be adopted for Standby smoothing reactor house. Reinforced concrete bent structure will be adopted for maintenance workshop & spare parts storage and its roof truss is the purlin roof structure system.

建维生产区建筑采用砌体结构，现浇钢筋混凝土楼（屋）面；

The building of Temporal Area adopts masonry structure and cast-in-situ reinforced concrete floor.

综合泵房及消防泵房地面以上采用钢筋混凝土框架结构，地面以下池体采用现浇钢筋混凝土箱型结构。

The integrated water pump house includes above-ground and underground parts. The above-ground part is in the single-floor reinforcement concrete frame structure and underground part will be in the form of pond and is in the cast-in-situ reinforced concrete box structure.

其余建筑物均采用钢筋混凝土框架结构，包括控制楼、继电器室、办公楼、中巴方食堂、宿舍、专用品库等。

Reinforcement concrete frame structure will be adopted for other buildings, including control building, relay and battery building, office buildings, refectories, refectories，specially store，metering buildings，common switchgear building and etc.

3.3.2 设备基础及构（支）架

换流变基础与防火墙基础采用钢筋混凝土桩筏板基础。防火墙采用钢筋混凝土框架+砌体填充墙结构。搬运轨道采用钢筋混凝土桩筏板基础。

3.3.2 Gantries and Equipment Supports

The flow change and firewall adopt reinforced concrete pile raft foundation. The firewall adopts reinforced concrete frame + masonry filling wall structure. Reinforced concrete pile raft foundation for moving track.

全站构架均采用门型架结构。构架柱采用矩形变截面角钢格构式塔架，挂线横梁采用矩

The whole station structure adopts a door frame structure. The frame column adopts a

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形角钢格构式钢梁，梁柱刚接。塔架与基础采用地脚螺栓连接。基础采用钢筋混凝土独立桩基承台。独立避雷线等塔架采用矩形变截面格构式角钢塔架，柱脚与基础采用地脚螺栓连接。基础采用钢筋混凝土独立桩基承台。

rectangular variable section steel lattice frame, and the pylon beam adopts a rectangular angle steel lattice steel beam, and the beam column is rigidly connected. The base and base are connected by anchor bolts. The foundation adopts reinforced concrete independent pile foundation support platform. The independent lightning rod tower adopts a rectangular variable section lattice type corner steel tower frame, and the column foot is connected with the foundation by a foot bolt. The foundation adopts reinforced concrete independent pile foundation support platform.

全站设备支架均采用矩形角钢格构式支架柱，与基础采用地脚螺栓连接。基础采用钢筋混凝土独立桩承台。

The support of the whole station equipment adopts rectangular steel lattice support column and is connected with the foundation by foot bolts. The foundation adopts reinforced concrete independent pile bearing platform.

交流滤波器组、平波电抗器等设备基础采用钢筋混凝土独立桩承台。

The equipment foundation of AC filter group and flat wave reactor is reinforced concrete independent pile bearing platform.

3.3.3 水工构筑物

地下的水工构筑物如消防水池、雨水泵站、工业水池、污水调节池等均采用地下钢筋混凝土箱型结构。

污水处理装置基础采用整板基础。

3.3.3 Hydraulic structures

Underground hydraulic structures such as Fire-fighting pools, rainwater pumping stations, industrial pools, and sewage adjustment pools all use underground reinforced concrete box structures.

Whole board foundation is used for sewage treatment plant foundation.

3.3.4 钢结构防腐处理

所有构（支）架、避雷线塔架等钢结构采用热镀锌工艺防腐。

钢结构屋架采用刷漆防腐（水性无机富锌底漆+环氧云铁中间漆+聚氨酯面漆）。

现场焊接的局部部位及镀锌层破坏处采用现场冷喷锌防腐。

3.3.4 Anti-corrosion of steel structure

The steel for gantry and equipment support will be hot-galvanized for corrosion resistance. The steel buildings will be painted for corrosion resistance.And zinc-rich painting will be used at site for local anti-corrosion treatment where the zinc coat is damaged.

4 地基处理

4.1 岩土地质

地基岩土层从上到下依次描述如下：

4 SUBGRADE AND FOUNDATION

4.1 Geological conditions

The rock and soil layers of foundation are described as follows from top to bottom:

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①层：粉质粘土，黄褐色、褐色，稍湿，以可塑状态为主，局部夹粉土，表层一般分布 30cm 厚耕植土。该层层位相对稳定，在整个场区广泛分布，厚度 2.0~7.0m，标贯击数 5.5 击，压缩模量 Es 值为 6.5MPa，地基承载力为 70kPa；

①Layer: Silty clay, tawny, brown, slightly wet, mainly in a plastic state, locally interbedded with silt. 30cm thick cultivated soil is generally distributed on the surface layer. The layer has a relatively stable position and is widely distributed in the entire site, with the thickness of 2.0~7.0m, standard penetration test(SPT) blow count of 5.5, the compression modulus Es of 6.5MPa and the foundation bearing capacity of 70kPa;

②层：粉细砂，灰色、青灰色，饱和，含有少量云母片，局部见砾石，夹少量粉土和中砂，分为②$_1$、②$_2$、②$_3$、②$_4$ 和②$_5$ 五个亚层；

②Layer: Silty-fine sand, gray, green gray, saturated, containing a small amount of mica, locally with gravel, with a small amount of silt and medium sand. The layer is divided into five sublayers as ②$_1$,②$_2$,②$_3$,②$_4$ and ②$_5$;

②$_1$ 粉细砂：松散，层厚 2.0～7.0m，标贯击数 7.8 击，压缩模量 Es 值为 5.0MPa，地基承载力为 80kPa，该层在拟建场地内广泛分布；

②$_1$silty-fine sand is loose, with layer thickness of 2.0~7.0m, SPT blow count of 7.8, compression modulus Es of 5.0MPa, and foundation bearing capacity of 80kPa. The layer is widely distributed in the site;

②$_2$ 粉细砂：稍密，层厚 3.0～7.0m，标贯击数 12.6 击，压缩模量 Es 值为 7.0MPa，地基承载力为 100kPa，该层在拟建场地内广泛分布；

②$_2$Silty-fine sand is slightly dense, with layer thickness of 3.0 ~ 7.0m, SPT blow count of 12.6, compression modulus Es of 7.0 MPa, and foundation bearing capacity of 100kPa. This layer is widely distributed in the proposed site;

②$_3$ 粉细砂：中密，层厚 6.5~16.0m，标贯击数 22.2 击，压缩模量 Es 值为 15.0MPa，该层在拟建场地内广泛分布；

②$_3$Silty-fine sand is medium dense, with layer thickness of 6.5 ~ 16.0m, SPT blow count of 22.2, and compression modulus Es of 15.0 MPa. This layer is widely distributed in the site;

②$_4$ 粉细砂：密实，层厚 10.0~18.5m，标贯击数 35.2 击，压缩模量 Es 值为 24.0MPa，该层在拟建场地内广泛分布；

②$_4$Silty-fine sand is dense, with layer thickness of 10.0 ~ 18.5m, SPT blow count of 35.2, and compression modulus Es of 24.0 MPa. This layer is widely distributed in the site;

②$_5$ 粉细砂：极密，标贯击数大于 50 击，压缩模量 Es 值为 30.0MPa，在站址区均有分布，本次勘测未揭穿该层。

②$_5$Silty-fine sand is very dense, with SPT blow count of larger than 50, and compression modulus Es of 30.0MPa. This layer is evenly distributed in the site and was not exposed in this survey.

本工程场地的地震基本烈度为 7 度，地面下存在饱和砂土和饱和粉土时，应进行液化判定。按较高地下水位埋深 2.0m 进行考虑，根据美标 Seed 法（SPT）计算结果，本工程场地土在 7 度地震时，②$_1$、②$_2$ 粉细砂可发生液化，

The converter station is located in the section of alluvial plain; there is saturated sand, with the risk of sandy soil liquefaction. The determination of this sandy soil liquefaction adopts American-standard of seed method

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液化没有明显的区域性，一般 12.0m 深度内的粉细砂可能发生液化，液化等级中等。

(SPT). from the calculation results, there is sandy soil liquefaction in ②$_1$、②$_2$ soil layer of this site.

此外，由于百年一遇洪水位影响，站址场地需外购土填高至百年一遇洪水位以上，填土厚度约 2.8m。

In addition, due to the impact of a flood level in a hundred years, the site site needs to purchase soil to fill up to a flood level in a hundred years, and the fill thickness is about 2.8 m.

4.2 地基处理方案

场地浅部地基土工程性质一般，强度较低、压缩性较高，其力学性能较差，且浅层 12 米范围内粉细砂土有存在中等液化情况，因此，对阀厅、控制楼、办公楼、检修备品库、换流变基础、交流滤波器组基础及构架、塔、支架等全站主要建构筑物均采用桩基进行处理，桩基采用机械成孔灌注桩。围墙、电缆沟、道路、各类箱体及灯具等次要构筑物采用天然地基，如局部遇不良地质，采用级配砂石换填处理。

4.2 Foundation Treatment Scheme

Generally, the engineering properties of the ground foundation soil are low strength, high compressibility, and its mechanical properties are poor, and the liquefaction of fine-silty in the shallow 12 meters, therefor pile foundation should be adopted in the main construction structures such as valve hall, control building, office building, towers and supportsand etc.and the cable trench, road, various types of boxes and lamps shall be built on undisturbed soil layer, otherwisethe graded sand should be used for replacement.

对于因百年一遇洪水位影响而需外购的回填土，要求采用分层碾压、逐层夯实，每次回填虚铺厚度不大于 300mm，最终压实系数不小于 0.94。

For the backfill soil that needs to be purchased due to the impact of the flood level for a hundred years, layered rolling and lamination are required. The thickness of each backfill virtual paving is not more than 300mm, and the final compaction coefficient is not less than 0.94.

4.3 边坡

由于站址购土抬高约 2.8m 后，在站址四周及进站道路两侧形成填方边坡。根据中国电力技术装备有限公司有关专题工作会纪要等，边坡采用自然放坡法，采用预制空心六棱块植草护坡，坡率按 1:1.5 考虑。坡脚处设置护脚墙防冲刷，坡面采用绿化处理。

4.3 Slope

Due to the elevation of about 2.8 m in the site purchase, a filled slope is formed around the site and on both sides of the entrance road. According to the minutes of the relevant special work meetings of China Electric Power Technology Equipment Co., Ltd., the slope adopts a natural slope method and adopts a prefabricated hollow hexagonal block to protect the slope. The slope rate is considered at 1:1.5. The foot guard wall is set up at the foot of the slope, and the slope surface is treated with greening.

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